

LucyTrace ☒

Turn "Ruff Drafts" into the "Director's Cut"



LucyTrace is a high-performance pipeline for converting raster images (PNG, JPG, WEBP, etc.) into clean vector SVGs and print-ready coloring pages.

Named after our Director of Wellness (Lucy the Cavachon), this tool ensures your images are well-groomed, free of noise, and ready for:

1. **Laser Cutting/Engraving:** Compatible with LightBurn, XTool Creative Space, and Glowforge.
2. **CAD/CNC:** Optimized node counts for smoother import into **Autodesk Fusion** sketches.
3. **Print:** Generates high-resolution, centered PDFs and PNGs for coloring books.

☒ Quick Start

1. **Drop your images** (PNG, JPG, BMP, etc.) into the root folder next to the script.
2. **Run the pipeline:**
 - **Windows:** Double-click `lucytrace.bat`.
 - **Mac/Linux:** Double-click `lucytrace.sh`.
3. **Choose your Profile** from the menu:
 - [1] **BASE** : Fast processing for clean line art.
 - [2] **ADVANCED** : Smart processing for photos, sketches, or complex images.
4. **Collect your files:**
 - `output_svg/` -> Vector files.
 - `output_png/` -> High-res raster files.
 - `output_pdf/` -> Print-ready documents.

⚙️ The Two Profiles

LucyTrace uses a unified engine but offers two distinct "grooming" styles:

1. Base Profile ("The Fast Track")

- **Best For:** High-quality scans, crisp digital line art, black ink on white paper.
- **Logic:** Applies a strict brightness threshold (default 60%). Pixels darker than the limit become black; everything else becomes white.
- **Pros:** Extremely fast and predictable on clean input.

2. Advanced Profile ("The Director's Cut")

- **Best For:** Photos of drawings (bad lighting), "muddy" scans, comic book panels, or inverted images (white lines on black).
- **Logic:**
 - **Auto-Levels:** Stretches contrast to fix lighting issues (makes dark grays black, light grays white).
 - **Smart Detection:** Automatically detects if an image is Color or B&W.
 - **Posterization:** If color is detected, it flattens gradients into chunks ("stained glass" style) before tracing to prevent noise.

☒ Configuration (`lucytrace.ini`)

You don't need to edit the Python code to change settings. All major toggles are in `lucytrace.ini` :

```
[General]
profile = adv          ; Default profile if running via CLI
overwrite = false     ; Set to true to force reprocessing existing files
export_width = 3000    ; Width of the final PNG (3000px is approx 10 inches at 300 DPI)

[ImageProcessing]
threshold_percent = 60% ; The cutoff for black vs white

[Modes]
invert = off           ; Set to 'on' if your input is white lines on a black background
mode = auto            ; Can force 'color' or 'bw' manually here
```

📁 Folder Structure

The project uses the following structure (folders are auto-created if missing):

```
<project folder>/
├─ input_images/    # Source images (auto-moved here from root)
├─ cleaned_png/     # Intermediate black/white PNGs (useful for debugging)
├─ output_svg/      # Traced vector output files (Optimized for CAD/Laser)
├─ output_png/      # Exported PNGs for printing
├─ output_pdf/      # Exported PDFs for printing
├─
├─ lucytrace.py     # The Master Script
├─ lucytrace.ini    # Configuration File
├─ lucytrace.bat    # Windows Launcher
├─ lucytrace.sh     # Mac/Linux Launcher
└─ README.md
```

📋 Requirements

LucyTrace requires **Python 3**, **ImageMagick**, **Potrace**, and **Inkscape**. All tools must be installed and accessible via your system PATH.

Installation Guide

🪟 Windows Setup

1. **Python 3**: Download from python.org. Ensure **"Add to PATH"** is checked.
2. **ImageMagick**: Download the DLL installer from imagemagick.org. Check **"Install legacy utilities"** and **"Add to PATH"**.
3. **Inkscape**: Install from inkscape.org.
4. **Potrace**:
 - Download the Windows zip from [SourceForge](https://sourceforge.net).
 - Extract it (e.g., to C:\Program Files\potrace).
 - Add that folder to your System Environment **PATH**.

🍏 macOS Setup

1. **Homebrew**: If you don't have it, install it first.
2. **Install Tools**:

```
brew install python imagemagick potrace --cask inkscape
```

3. **Link Inkscape**: macOS needs this link to find the tool:

```
sudo ln -s /Applications/Inkscape.app/Contents/MacOS/inkscape /usr/local/bin/inkscape
```

4. **Permissions**: Make the launcher executable:

```
chmod +x lucytrace.sh
```

🐧 Linux Setup (Debian/Ubuntu)

```
sudo apt update  
sudo apt install python3 python3-pip imagemagick potrace inkscape
```

Happy Making! ☺